

DATA UNDERSTANDING

El dataset recoge información demográfica, financiera y del historial de contacto de clientes de Banco Atlas. El objetivo es identificar qué características están asociadas con una mayor probabilidad de contratar un depósito a plazo tras una campaña de marketing.

Se detectan valores nulos en age, marital y education. El porcentaje es muy bajo, por lo que podrían eliminarse en la fase de limpieza sin afectar el dataset .

La variable objetivo (deposit) presenta una distribución equilibrada entre clientes que contrataron el depósito y clientes que no lo contrataron (49,7 % - 50,3 % aprox).

La población del dataset está compuesta principalmente por adultos de mediana edad.

La variable Balance presenta una fuerte asimetría positiva y posibles valores extremos, aunque estos pueden corresponder a clientes con patrimonio alto.

Duration sesgada a la derecha: la mayoría de las llamadas dura poco

- 10.642 registros
- 18 variables
- 1 registro = 1 cliente contactado

IDENTIFICADOR = id

DEMOGRÁFICAS = age, job, marital, education

FINANCIERAS = balance, default, housing, loan

CAMPAÑA = contact, day, month, duration, campaign

HISTORIAL = pdays, previous, poutcome

OBJETIVO = deposit

NaN:

- age: 9
- marital: 5
- education: 7

DESCRIBE():

BALANCE □ min -6847 / max 81204 si? No?

CAMPAIGN □ max = 63 (number of contacts performed during this campaign and for this client ??) Correcto? Y si si, tiene sentido a nivel comercial contactar tantas veces?

DURATION □ max = 64 min correcto?

PDAYS □ -1 = nunca contactado antes

UNKNOWN VALUES:

Poutcame = 7885

Contact = 2196

Education = 473

Job = 68

POUTCOME = **UNKNOWN 7885**

FAILURE 1180

SUCCESS 1065

OTHER 512

DEPOSIT = YES / NO 50% equilibrado

CONTACT = cellular 7707

unknown 2196

telephone 739

MONTH = max may 2646 / min December 109

Variables Table



Variable Name	Role	Type	Demographic	Description
age	Feature	Integer	Age	
job	Feature	Categorical	Occupation	type of job (categorical: 'admin.','blue-collar','entrepreneur','housemaid','mana employed','services','student','technician','unemployed','unknown')
marital	Feature	Categorical	Marital Status	marital status (categorical: 'divorced','married','single','unknown'; note: 'divorce widowed')
education	Feature	Categorical	Education Level	(categorical: 'basic.4y','basic.6y','basic.9y','high.school','illiterate','professional.course','univer
default	Feature	Binary		has credit in default?
balance	Feature	Integer		average yearly balance
housing	Feature	Binary		has housing loan?
loan	Feature	Binary		has personal loan?
contact	Feature	Categorical		contact communication type (categorical: 'cellular','telephone')

day_of_week	Feature	Date		last contact day of the week
month	Feature	Date		last contact month of year (categorical: 'jan', 'feb', 'mar', ..., 'nov', 'dec')
duration	Feature	Integer		last contact duration, in seconds (numeric). Important note: this attribute highl target (e.g., if duration=0 then y='no'). Yet, the duration is not known before a after the end of the call y is obviously known. Thus, this input should only be ir purposes and should be discarded if the intention is to have a realistic predicti
campaign	Feature	Integer		number of contacts performed during this campaign and for this client (numer contact)
pdays	Feature	Integer		number of days that passed by after the client was last contacted from a previo (numeric; -1 means client was not previously contacted)
previous	Feature	Integer		number of contacts performed before this campaign and for this client
poutcome	Feature	Categorical		outcome of the previous marketing campaign (categorical: 'failure','nonexisten
y	Target	Binary		has the client subscribed a term deposit?

